WELLSLING · MARKET AND VIABILITY

WellSling market and viability

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

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0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade)

WellSling market and viability

WellSling is for villages, dairy cooperatives and animal rescue groups in regions with many open wells, first in north India, who today either send a person down or use people-rated or improvised gear. No product found does the same job: confined-space tripods are rated for people and cost several thousand dollars, and farm cattle lifts assume the animal is lying on open ground, so an open design that a district workshop can build and a cooperative can share fills a real gap. Prices and benchmarks are cited with the date they were checked. Method: STANDARDS section 23.

> **Safety:** A low price must never come from a weaker rig. Every kit built from this design is proof-loaded at 1.5 times the hook load by a competent person before use.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Village and dairy cooperative kit	Farmers, volunteers and cooperative staff in villages with open wells; first north India (Haryana, Bihar, Uttar Pradesh)	Confined-space tripod with retrieval winch, rated for people (191 kg class): about USD 5,850 to 6,120 (3M DBI-SALA with Salalift II, PK Safety, as cited in WSL-PRB-001 on 2026-10-03; to recheck)	3,093 (BOM estimate)	Rated for a 600 kg animal, worked from the rim, built and repaired locally, shared between villages
Animal rescue organisations	Rescue vans covering many villages	Technical large-animal rescue kits (Becker and Anderson slings, bipods, haul systems) listed by the University of Florida veterinary response team; price to assess	3,093	Purpose-made for wells; the landing bridge avoids swinging the animal over the rim
Rural fire and disaster crews	District fire stations answering animal-in-well calls	People-rated tripods and improvised rope lifts (as above)	3,093	A non-entry method that keeps the crew out of the well
Farms with downed cattle (adjacent, not the target)	Dairy farms lifting a cow that is down in a shed or field	Cow Lifter Liftease sling accessory for a silage-fork lift: AUD 469 (OzFarmer, checked 2026-10-25; listed as sold out)	Not applicable	Shows farmers pay for gentle lifting slings; not a well rescue product

Option B's main bought item, for comparison: a 1 t hand chain pulley block with a 3 m lift is listed at INR 7,920 (Kismat Machines on IndiaMART, checked 2026-10-25); the 15 m lift option B needs is priced higher and is to assess.

Open route

- **Who builds it.** A district fabrication workshop with a welder, a drill press and access to a galvanising plant makes the tripod, feet, spreader, yokes and bridge from the drawings and the build plan; the winch, rope, rigging, slings and harnesses are bought to their standards (WSL-REG-001). The first kit would be built with the co-design partner at TRL 4.

- **How it spreads.** Through dairy cooperative unions and animal rescue organisations that already answer well calls, and through veterinary universities and district fire services. One kit per cluster of villages or per rescue van spreads the cost; a cooperative buying winches and rigging for several kits lowers it.
- **What keeps it open.** Hardware under CERN-OHL-S-2.0 and software under MIT (REUSE compliant), plain webbing bands and a separate passer to stay clear of the patents noted in WSL-PRC-001, and documents written for a capable maker with no special tools.
- **Viability.** No commercial model is proposed. Cost is reported against the USD 1,500 value-engineering target (USD 1,593 over); the savings worth trying are in WSL-DEC-001.

Sources

- OzFarmer. "Cow Lifter Liftease." Accessed October 25, 2026. https://ozfarmer.com/products/cow-lifter-liftease.
- IndiaMART. "Pulley Chain Block." Accessed October 25, 2026. https://m.indiamart.com/impcat/chain-blocks.html.
- PK Safety. "3M DBI-SALA Confined Space Aluminum Tripod with Winch." As cited in WSL-PRB-001 (2026-10-03). https://pksafety.com/products/3m-dbi-sala-confined-space-aluminum-tripod-w-winch-83010.
- University of Florida College of Veterinary Medicine, Veterinary Emergency Treatment Service. "ATR Equipment Resources." Accessed October 25, 2026. https://responseteam.vetmed.ufl.edu/animal-technical-rescue/atr-equipment-resources/.